

Tapas Pal

Group Leader | Secure Computation & Communication, KASTEL SRL,
Karlsruhe Institute of Technology (KIT)

Ph.D. in Cryptography, IIT Kharagpur

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🎓 Google Scholar DBLP IACR CryptoDB



RESEARCH PROFILE

Cryptographer working on the foundations and real-world deployment of advanced public-key encryption. My research designs provably secure *functional*, *attribute-based*, and *registered* encryption schemes that decentralize trust, make computation on encrypted data *verifiable*, and bring strong privacy guarantees to cloud platforms, digital infrastructure, and AI systems. The work spans rigorous theory (pairing- and lattice-based constructions with formal security proofs) and applied impact (multiple international patents and industry technology due-diligence). Author of 20+ peer-reviewed papers at leading venues (CRYPTO, EUROCRYPT, ASIACRYPT, PKC, Journal of Cryptology).

Research Interests Functional & attribute-based encryption; registered and multi-authority encryption; verifiable computation on encrypted data; privacy-enhancing technologies; lattice- and pairing-based cryptography; witness encryption & obfuscation; threshold and post-quantum cryptography.

EDUCATION

Ph.D. in Cryptography, Indian Institute of Technology Kharagpur, India 2016 – 2021
Advisor: Prof. Ratna Dutta. Thesis: *Designing Provably Secure Advanced Cryptographic Primitives — Witness Encryption, Fully Homomorphic Encryption, and Functional Encryption*. Defended 6 October 2021.

M.Sc. in Mathematics, Indian Institute of Technology Kharagpur, India 2014 – 2016
CGPA 9.47/10 · Batch Rank 2.

B.Sc. (Honours) in Mathematics, Scottish Church College, University of Calcutta, India 2011 – 2014
First Class First · 88.88%.

ACADEMIC & RESEARCH APPOINTMENTS

Group Leader, Secure Computation & Communication, KASTEL Security Research Labs, KIT, Germany 2024 – present
Lead a research group on privacy-enhancing technologies; research strategy, proposal coordination, and mentoring of early-career researchers.

Postdoctoral Fellow, Social Informatics Laboratories, NTT Corporation, Tokyo, Japan 2022 – 2023
Computing on encrypted data and advanced functional-encryption primitives.

Visiting Researcher, Cyber Security Group, University of St. Gallen, Switzerland 2023: Sep-Oct
Supervised the project on multi-client attribute-based unbounded inner-product functional encryption.

Visiting Scientist, R. C. Bose Centre for Cryptology & Security, ISI Kolkata, India 2021: Nov-Dec
Functional encryption and its applications.

Research Intern, Cryptography & Information Security (CIS) Lab, NTT Research, Inc., Sunnyvale, USA 2021: Feb-Oct
Functional encryption.

Junior & Senior Research Fellow, Department of Mathematics, IIT Kharagpur, India 2016 – 2021
Doctoral research in cryptography.

RESEARCH FUNDING & GRANTS

Swiss National Science Foundation (SNSF), University of St. Gallen 2023
Multi-Client Unbounded Attribute-Based Inner Product Functional Encryption — with PI Prof. Katerina Mitrokotsa. (Completed.)

SELECTED HONOURS & AWARDS

Competing position offers (declined) Senior Researcher, Cyber Security Group, University of St. Gallen, Switzerland; and Senior Scientist / Lecturer, Cyber Security Research Group, University of Klagenfurt, Austria.	2023
ERCIM Postdoctoral Fellowship , NTNU, Norway European Research Consortium for Informatics and Mathematics.	2021
Junior Research Fellowship (JRF), UGC , Mathematical Sciences All-India Rank 78.	2016
Joint Admission Test for M.Sc. (JAM) , Mathematics All-India Rank 121.	2014
Seven merit medals & prizes for excellence in Mathematics , Scottish Church College Including the Krishnalal De Medal (best performance in Mathematics), Pravat Kr. Ghosh Medal, Prof. Alexander Thomson Prize, and Dr. Alexander Duff Memorial Prize.	2013 – 2014

PUBLICATIONS

In theoretical cryptography, author names are listed *alphabetically* by convention; my name is shown in **bold**. Full list: [Google Scholar](#) · [DBLP](#) · [IACR CryptoDB](#).

Bibliometrics: Citations 205 · h-index 09 (Google Scholar, June 7, 2026).

Peer-Reviewed Conference Proceedings

- [C1] **CRYPTO 2026**. Shalini Banerjee, **Tapas Pal**, Andy Rupp, Daniel Slamanig. *Simple Public-Key Anamorphic Encryption and Signature using Multi-Message Extensions*. (To appear.) [\[ePrint\]](#)
- [C2] **PKC 2026**. **Tapas Pal**, Robert Schädlich. *Registered Functional Encryption for Attribute-Weighted Sums with Access Control*. 29th Intl. Conf. on Practice and Theory of Public-Key Cryptography. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C3] **ASIACRYPT 2025**. **Tapas Pal**, Robert Schädlich. *A General Framework for Registered Functional Encryption via User-Specific Pre-Constraining*. Advances in Cryptology — ASIACRYPT 2025. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C4] **PKC 2025**. Subhranil Dutta, Aikaterini Mitrokotsa, **Tapas Pal**, Tomy Jenit. *Multi-Client Attribute-Based Unbounded Inner Product Functional Encryption, and More*. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C5] **ASIACRYPT 2024**. Pratish Datta, **Tapas Pal**, Shota Yamada. *Registered FE beyond Predicates: (Attribute-Based) Linear Functions and More*. Advances in Cryptology — ASIACRYPT 2024. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C6] **IEEE EuroS&P 2024**. Uddipana Dowerah, Subhranil Dutta, Frank Hartmann, Aikaterini Mitrokotsa, Sayantan Mukherjee, **Tapas Pal**. *SACfe: Secure Access Control in Functional Encryption with Unbounded Data*. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C7] **EUROCRYPT 2024**. Taiga Hiroka, Fuyuki Kitagawa, Tomoyuki Morimae, Ryo Nishimaki, **Tapas Pal**, Takashi Yamakawa. *Certified Everlasting Secure Collusion-Resistant Functional Encryption, and More*. Advances in Cryptology — EUROCRYPT 2024. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C8] **PKC 2023**. Pratish Datta, **Tapas Pal**. *Decentralized Multi-Authority Attribute-Based Inner-Product FE: Large Universe and Unbounded*. 26th Intl. Conf. on Practice and Theory of Public-Key Cryptography. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C9] **ASIACRYPT 2022**. Pratish Datta, **Tapas Pal**, Katsuyuki Takashima. *Compact FE for Unbounded Attribute-Weighted Sums for Logspace from SXDH*. Advances in Cryptology — ASIACRYPT 2022. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C10] **ASIACRYPT 2021**. Pratish Datta, **Tapas Pal**. *Compact Adaptively Secure FE for Attribute-Weighted Sums from k -Lin*. Advances in Cryptology — ASIACRYPT 2021. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C11] **ProvSec 2021**. Subhranil Dutta, **Tapas Pal**, Ratna Dutta. *Fully Secure Unbounded Zero Inner Product Encryption with Short Ciphertexts and Keys*. 15th Intl. Conf. on Provable and Practical Security. [\[Proceedings\]](#)
- [C12] **ACISP 2021**. **Tapas Pal**, Ratna Dutta. *Chosen-Ciphertext Secure Functional Encryption from Constrained Witness PRF*. 26th Australasian Conf. on Information Security and Privacy. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C13] **ACISP 2021**. **Tapas Pal**, Ratna Dutta. *CCA-Secure Attribute-Hiding Inner Product Encryption from Minimal Assumption*. 26th Australasian Conf. on Information Security and Privacy. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C14] **LATINCRYPT 2021**. **Tapas Pal**, Ratna Dutta. *Attribute-Based Access Control for Inner Product Functional Encryption from LWE*. 7th Intl. Conf. on Cryptology and Information Security in Latin America. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C15] **CANS 2020**. **Tapas Pal**, Ratna Dutta. *Chosen-Ciphertext Secure Multi-Identity and Multi-Attribute Pure FHE*. 19th Intl. Conf. on Cryptology and Network Security. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C16] **ProvSec 2020**. **Tapas Pal**, Ratna Dutta. *Semi-Adaptively Secure Offline Witness Encryption from Puncturable Witness PRF*. 14th Intl. Conf. on Provable and Practical Security. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C17] **ACISP 2019**. **Tapas Pal**, Ratna Dutta. *Offline Witness Encryption from Witness PRF and Randomized Encoding in CRS Model*. 24th Australasian Conf. on Information Security and Privacy. [\[Proceedings\]](#) [\[ePrint\]](#)

Journal Articles

- [J1] **IACR CiC 2025**. Subhranil Dutta, **Tapas Pal**, Amit Kumar Singh, Sourav Mukhopadhyay. *Fully Collusion Resistant Traceable Identity-Based Inner Product Functional Encryption*. IACR Communications in Cryptology. [\[Journal\]](#)

- [J2] **DCC 2024**. Pratish Datta, **Tapas Pal**, Katsuyuki Takashima. *Compact FE for Unbounded Attribute-Weighted Sums for Logspace from SXDH*. Designs, Codes and Cryptography. [Journal]
- [J3] **TCS 2024**. Subhranil Dutta, **Tapas Pal**, Ratna Dutta. *Reinforcing Privacy in Cloud Computing via Adaptively Secure Non-Zero Inner Product Encryption and Anonymous Identity-Based Revocation in the Unbounded Setting*. Theoretical Computer Science, vol. 995. [Journal]
- [J4] **JoC 2023**. Uddipana Dowerah, Subhranil Dutta, Aikaterini Mitrokotsa, Sayantan Mukherjee, **Tapas Pal**. *Unbounded Predicate Inner Product Functional Encryption from Pairings*. Journal of Cryptology, 36(29). [Journal] [ePrint]
- [J5] **DCC 2023**. Pratish Datta, **Tapas Pal**. *Compact Adaptively Secure FE for Attribute-Weighted Sums from k -Lin*. Designs, Codes and Cryptography, 91, 2917–3034. [Journal] [ePrint]

Manuscripts under Submission / Preprints

- [P1] **Preprint 2025**. **Tapas Pal**, Robert Schädlich, Erkan Tairi. *Registered Functional Encryption for Pseudorandom Functionalities from Lattices: Registered ABE for Unbounded-Depth Circuits and Turing Machines, and More*. [ePrint]

PATENTS

- [PT1] **US 2024**. Pratish Datta, **Tapas Pal**. *Decentralized Multi-Authority Attribute-Based Encryption for Large Universe and Unbounded*. US Patent WO2024151871A1. [Google Patents]
- [PT2] **US 2024**. Pratish Datta, **Tapas Pal**. *Compact Functional Encryption for Unbounded Attribute-Weighted Sums*. US Patent WO2024098074A2. [Google Patents]
- [PT3] **US/JP/EPO 2023**. Pratish Datta, Monosij Maitra, **Tapas Pal**. *Decentralized Multi-Authority Attribute-Based Inner-Product Functional Encryption*. Patent EP4254858A1. [Google Patents]
- [PT4] **US 2023**. Pratish Datta, **Tapas Pal**. *Compact Adaptively Secure Functional Encryption for Attribute-Weighted Sums*. US Patent WO2023014969A1. [Google Patents]

INVITED TALKS & RESEARCH VISITS

Invited Talks

- Swiss Crypto Day**, ETH Zurich, Switzerland Sep 2023
Inner Product Functional Encryption for the Real World.
- Research Seminar**, University of Auckland, New Zealand Jun 2019
From Functional Encryption to Indistinguishability Obfuscation.

Research Visits

- University of St. Gallen, Switzerland**, Cyber Security Group 2023
Host: Prof. Katerina Mitrokotsa.
- IIT Madras, India** 2019
Host: Prof. Shweta Agrawal.
- University of Auckland, New Zealand** 2019
Host: Prof. Steven D. Galbraith.

TEACHING EXPERIENCE

- Cryptography and Network Security — Teaching Assistant**, NPTEL / IIT Kharagpur 2018 – 2020
Spring 2018, 2019, 2020.
- Introduction to Abstract and Linear Algebra — Teaching Assistant**, NPTEL / IIT Kharagpur 2018 – 2020
Autumn 2018, 2019, 2020.
- Mathematics-II — Teaching Assistant**, IIT Kharagpur 2019
Spring 2019.

PROFESSIONAL SERVICE

- Program Committee** PKC 2027; ACM CCS 2024, 2026; WWW 2026; INDOCRYPT 2024, 2025; ICSP 2024, 2025.
- External Reviewer** CRYPTO 2022–2026; EUROCRYPT 2024–2026; ASIACRYPT 2024, 2025; TCC 2022, 2025; PKC 2023, 2024; ACNS 2022; AsiaCCS 2022; INDOCRYPT 2020, 2021.
- Journal Referee** Journal of Cryptographic Engineering; Designs, Codes and Cryptography; IEEE Trans. Information Forensics & Security; IEEE Trans. Information Theory; IET Information Security; Advances in Mathematics of Communications; Computer Standards & Interfaces.
- Expert Reviewer** American Mathematical Society (AMS): Mathematical Reviews Database

INDUSTRY CONSULTING & KNOWLEDGE TRANSFER

Cryptography Consultant (pro bono) , D11Z.Ventures GmbH & Co. KG	Nov 2025
Technical due-diligence on a password-less access-control platform (MyCena); startup evaluation and advisory.	
Cryptography Consultant (pro bono) , D11Z.Ventures GmbH & Co. KG	Mar 2025
Technical due-diligence on cryptographic image-authenticity technology (TrustNXT); startup evaluation and advisory.	

TECHNICAL SKILLS & LANGUAGES

Programming	C/C++.
Scientific Computing	MATLAB, Mathematica.
Tools & Typesetting	$\text{\LaTeX}$, Linux.
Languages	Bengali (native); Hindi (fluent); English (fluent, full professional working proficiency).

REFERENCES

Prof. Ratna Dutta

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Dr. Andy Rupp

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